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Week 05: Monday, AST 5011: Astrophysical Systems

Stellar and Exoplanet Spectra

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With contributions totally ripped off from Carl Fields (UA), Mike Zingale (SUNY), Cole Miller (UMD), and Abi Nolan (Purdue).

Spectral Types (Spectral Classes)

See Appendix A of HKT for more details.

In Lectures 7 and 8, we explored the interior physics of stars: nuclear energy generation through the pp chain and CNO cycle, energy transport by radiation and convection, and the endpoints of stellar evolution from white dwarfs to neutron stars. Now we turn to the stellar surface. Spectral classification is the primary observational tool for measuring stellar temperatures and luminosities, providing the empirical foundation for everything we have discussed so far.

Stellar spectral types are based on the Henry Draper (HD) classification scheme, which classifies stars according to the appearance and strength of spectral lines from hydrogen, helium, and various metals.

The modern spectral classification scheme was developed at Harvard Observatory in the early 20th century. The work began with Henry Draper, who photographed the first stellar spectrum (Vega) in 1872. After Draper's death, his wife funded the continuation of the project, providing both equipment and financial support.

The bulk of the classification work was carried out by Annie Jump Cannon between 1918 and 1924. The original scheme used alphabetically ordered capital letters, but subsequent revisions reordered and condensed the sequence as stellar physics and evolution became better understood. The results were published in the Henry Draper Catalogue (HD) and Henry Draper Extension (HDE), containing spectra for over 225,000 stars down to ninth magnitude.

 AJC

The International Astronomical Union later adopted Cannon's method as the official spectral classification system. In addition to her scientific contributions, Cannon was also an advocate for women's rights and an active member of the National Women's Party.

Spectral Class Characteristics

The spectral sequence is fundamentally a temperature sequence, not a composition sequence. Nearly all main-sequence stars have similar bulk compositions (roughly 70% hydrogen by mass), but different effective temperatures produce different ionization and excitation states. The physics is governed by the Saha ionization equation and the Boltzmann excitation distribution: at a given temperature, each element has a specific fraction in each ionization stage and excitation level, and it is this balance that determines which spectral lines are strong.

O stars show strong He II lines, indicating very high effective temperatures ($T_{\text{eff}} \gtrsim 30,000$ K). These are the most massive main-sequence stars, burning hydrogen predominantly through the CNO cycle discussed in Lecture 7.

B stars show strong He I lines along with some hydrogen lines.

From A to F stars, the hydrogen Balmer lines reach their maximum strength in A stars (near $T_{\text{eff}} \approx 10,000$ K) and then weaken toward F. The peak occurs because the Boltzmann factor optimally populates the $n = 2$ level of hydrogen at this temperature. At higher temperatures hydrogen ionizes, removing the Balmer absorber; at lower temperatures the $n = 2$ level is not thermally populated.

G stars show strong calcium H and K lines. The Sun is a G2 star with $T_{\text{eff}} \approx 5800$ K, burning hydrogen through the pp chain.

K stars show increasingly strong metallic absorption lines as the temperature drops further.

M stars are dominated by molecular bands, especially TiO, because temperatures are low enough for molecules to survive in the photosphere.

L stars show strong metal-hydride bands such as CrH and FeH, while TiO and VO are largely absent. L dwarfs straddle the boundary between the lowest-mass hydrogen-burning stars and brown dwarfs.

T stars show strong methane (CH_4) absorption, similar to the spectrum of Jupiter. These are cool brown dwarfs with $T_{\text{eff}} \lesssim 1300$ K.

Effective temperature decreases in the sequence:

O $\rightarrow$ B $\rightarrow$ A $\rightarrow$ F $\rightarrow$ G $\rightarrow$ K $\rightarrow$ M $\rightarrow$ L $\rightarrow$ T

Each spectral class is subdivided numerically (e.g., G0, G1, G2) in order of decreasing temperature. The Sun is a G2 star.



Spectra of main-sequence (luminosity class V) stars of spectral classes O–M, adapted from Silva and Cornell (1992).

Electromagnetic Wavelength Bands

Name	Wavelength interval
Extreme ultraviolet	$< 1000 \text{ \AA}$
Ultraviolet	$1000\text{--}3000 \text{ \AA}$
Classical	$3000\text{--}4900 \text{ \AA}$
Visual	$4900\text{--}7000 \text{ \AA}$
Near infrared	$7000\text{--}10^5 \text{ \AA}$
Far infrared	$1\text{--}10^3 \mu\text{m}$
Radio	$0.1\text{--}10^4 \text{ cm}$

Luminosity Classes

The Morgan–Keenan (MK) classification combines spectral class and luminosity class, providing a two-dimensional description of stellar properties that closely corresponds to positions on the HR diagram.

The physical basis for luminosity class is surface gravity. At a given effective temperature, a supergiant has much lower surface gravity than a dwarf. Lower gravity means lower atmospheric pressure, which reduces collisional (pressure) broadening of spectral lines. Spectroscopists can therefore distinguish a supergiant from a dwarf of the same spectral type by examining line widths alone.

Luminosity class I stars are the most intrinsically luminous and are known as supergiants.

Luminosity class II stars are bright giants, less luminous than class I.

The numbering continues down to luminosity class V, which corresponds to dwarfs, or main-sequence stars.

The Sun is a G2V star. An M3I star is cool, extremely large, and intrinsically very luminous.

Compact remnants also have spectral classifications. White dwarfs (Lecture 8) are designated DA (hydrogen-dominated atmosphere), DB (helium-dominated), DC (featureless continuum), and other subtypes, depending on which elements are visible in their spectra. Since white dwarfs no longer undergo nuclear burning, their atmospheric composition reflects their prior evolutionary history and the settling of heavier elements under strong surface gravity.

In-Class Exercise: Spectral Type from Line Strengths

Goal

Use spectral line strengths to infer stellar temperature and spectral class, emphasizing excitation and ionization rather than composition.

Task

You are given a simplified dataset representing three stars. For each star, the equivalent widths of three spectral features are provided:

- H α (hydrogen Balmer line)
- He I
- Ca II K

Assume all stars have roughly solar composition.

1. Rank the stars from hottest to coolest.
2. Assign a rough spectral class (O, B, A, F, G, K, or M).
3. Explain your reasoning in terms of excitation and ionization.

Data

```
import numpy as np
```

```
stars = {  
    "Star A": {"H_alpha": 12.0, "He_I": 0.2, "Ca_K": 0.1},  
    "Star B": {"H_alpha": 3.0, "He_I": 5.5, "Ca_K": 0.05},  
    "Star C": {"H_alpha": 1.5, "He_I": 0.0, "Ca_K": 6.0},  
}
```

► Solution

Line Profiles and the Curve of Growth

In Lecture 7, we discussed opacity as the key quantity governing radiative energy transport inside stars. Here we examine how opacity manifests at the stellar surface through spectral line profiles. The same atomic physics that determines interior opacities (bound-bound, bound-free, and free-free transitions) also produces the absorption lines we use to classify and characterize stars.

Absorption of radiation by atoms can be described using a classical oscillator model.

For stationary atoms, the absorption line has a Lorentzian profile with a characteristic natural width γ set by the finite lifetime of the excited state (the Heisenberg uncertainty principle gives $\Delta E \cdot \Delta t \sim \hbar$). When atomic motions are included, Doppler shifts broaden the line into a Gaussian core. The resulting combined profile is called a Voigt profile.

 sigma_a

Shown is the absorption cross section as temperature increases from a pure Lorentzian profile ($\delta\nu_D = 0$) to Doppler-broadened lines. The natural width is set such that $\gamma/4\pi = 1$.

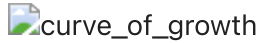
Here ν_0 is the resonant photon frequency, and γ is the sum of the damping constants of the two energy levels involved.

In real stellar atmospheres, additional broadening mechanisms operate. Pressure broadening (also called collisional broadening) arises from interactions with neighboring particles and dominates in high-density environments like dwarf star atmospheres. This is why luminosity class can be inferred from line widths: supergiants have lower atmospheric pressures and therefore narrower pressure-broadened lines than dwarfs.

Doppler Width

The Doppler width is the half-width at half-maximum of the line when Doppler broadening dominates over the natural width. It depends on temperature as $\Delta\nu_D \propto \sqrt{T/m}$, where m is the atomic mass, so heavier elements produce narrower Doppler cores at the same temperature.

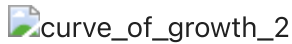
Curve of Growth



The curve of growth shows how the strength of an absorption line evolves as the number density of absorbing atoms increases. It is a fundamental diagnostic tool because it connects observable line strengths to physical conditions (abundances and temperatures) in the stellar atmosphere.

In practice, the equivalent width is measured as

$$W_\nu = \int_0^\infty \left(1 - \frac{I_\nu}{I_0}\right) d\nu = \int_0^\infty \left(1 - e^{-\beta_0}\right) d\nu$$



The curve of growth has three distinct regimes:

For small β_0 , corresponding to weak lines, $W_\nu \approx \beta_0$. This is the linear regime, where the equivalent width grows in direct proportion to the number of absorbers.

In the saturated regime, the line core has reached maximum depth (the intensity at line center is nearly zero). Adding more absorbers cannot deepen the core further, so W_ν scales approximately as $\sqrt{\ln \beta_0}$ and the curve rises very slowly. This flat portion makes abundance determinations from saturated lines unreliable.

For large β_0 , the damping wings dominate and $W_\nu \propto \beta_0^{1/2}$. Growth resumes because the Lorentzian wings extend far from line center and continue to absorb even after the core is fully saturated.

In-Class Exercise: Building a Curve of Growth Numerically

Goal

Develop intuition for the curve of growth by computing equivalent widths from a simple absorption model.

Task

We model an absorption line with a Gaussian optical depth profile:

$$\tau(\nu) = \tau_0 \exp\left[-\frac{(\nu-\nu_0)^2}{2\sigma^2}\right]$$

The observed intensity is

$$I(\nu) = I_0 e^{-\tau(\nu)}$$

1. Compute the equivalent width W_ν for different values of τ_0 .
2. Plot W_ν versus τ_0 on log-log axes.
3. Identify the linear and saturated regimes.

Conceptual questions

- Why does the curve flatten at intermediate optical depth?
- What physics is missing compared to a real stellar absorption line?
- How would damping wings change the curve?

► Solution

Exoplanets, Brown Dwarfs, and Stars

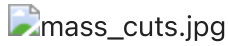
The distinction between stars, brown dwarfs, and planets is fundamentally about nuclear burning thresholds. In Lecture 7, we saw that thermonuclear reaction rates depend exponentially on temperature through the Gamow peak: the probability of tunneling through the Coulomb barrier rises steeply with temperature. This sets sharp mass boundaries for different types of nuclear burning.

Stars undergo sustained nuclear fusion of hydrogen in their cores. The minimum mass for hydrogen burning is approximately $0.085 M_\odot$, corresponding to a core temperature sufficient to sustain pp chain reactions (Lecture 7).

Brown dwarfs are not massive enough to sustain hydrogen fusion, with masses $M \lesssim 0.085 M_\odot$. Hydrogen burning requires core temperatures of approximately $T \sim 10^7$ K. Although brown dwarfs

cannot fuse hydrogen to helium, they can burn deuterium (^2H) at lower temperatures of $T \sim 10^6$ K. Deuterium has a lower Coulomb barrier than ordinary hydrogen (the reaction $^2\text{H} + p$ has a smaller charge product than $p + p$), so the Gamow peak occurs at lower temperature.

Planets undergo neither hydrogen nor deuterium fusion and occupy the mass range $M \lesssim 0.05 M_{\odot}$.



Credit: HKT Fig. 2.4. Mass cut diagram showing the fate of single objects in different mass regimes.

Planet Formation

More details can be found in the chapter by Phil Armitage in the Handbook of Exoplanets.

Planet formation is described by a bottom-up process in which planets form within gaseous protoplanetary disks from initially microscopic solid material. In Lecture 8, we discussed how molecular clouds collapse under the Jeans instability to form protostars. The same collapsing cloud also produces a circumstellar disk, and it is within this disk that planet formation occurs. Different physical processes dominate as growth proceeds.

1. Early growth phases involve particle sizes from microns to meters and are dominated by aerodynamic effects and material physics. Dust grains collide and stick, gradually building larger aggregates.
2. Gravitational interactions become important at later stages, first between planetesimals with sizes $\gtrsim \text{km}$, and later between growing protoplanets and the gas disk for masses $M \gtrsim 0.1 M_{\oplus}$.
3. Giant planet formation proceeds once cores reach masses of roughly $3\text{--}20 M_{\oplus}$. Growth is initially limited by the cooling of the gaseous envelope and later by the rate at which the disk can supply gas. This must happen before the gas disk disperses, typically within a few million years, setting a deadline for giant planet formation.

Transit Photometry as an Exoplanet Discovery Method



The figure shows the fraction of known exoplanets discovered by different detection methods as a function of time since 1995. At the end of 1995, only five planets were known, discovered via pulsar timing and radial velocity methods. From 1996 through roughly 2013, radial velocity discoveries dominated. By 2018, approximately 78% of known planets had been discovered using transit photometry.

The data are based on the NASA Exoplanet Archive as of February 2018. Timing methods include pulsar timing, eclipse timing, and transit timing. Other techniques such as astrometry and orbital brightness variations contribute only a small fraction of detections.

As a planet passes in front of its host star, the observed stellar flux decreases by a fractional amount ΔF . Assuming negligible planetary flux and spherical star and planet geometries, the transit depth is given by the ratio of their projected areas,

$$\Delta F = \left(\frac{R_{\text{planet}}}{R_{\text{star}}} \right)^2 \equiv k^2.$$

The total transit duration is denoted t_T , while t_F is the duration of full transit when the planet disk is entirely in front of the stellar disk.

These quantities allow the definition of the impact parameter,

$$b = \frac{a}{R_{\text{star}}} \cos i,$$

which can also be expressed in terms of observables as

$$b = \left[\frac{(1 - k)^2 - (\sin^2(\pi t_F / P) / \sin^2(\pi t_T / P)) (1 + k)^2}{\cos^2(\pi t_F / P) / \cos^2(\pi t_T / P)} \right]^{1/2}.$$

Here a is the orbital semimajor axis, i is the orbital inclination, and P is the orbital period.

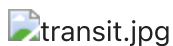
The system scale can be expressed as

$$\frac{a}{R_{\text{star}}} = \frac{1}{\sqrt{\tan(\pi t_T / P)}} \sqrt{(1 + k)^2 - b^2}.$$

Combining this with Kepler's laws allows an estimate of the stellar mean density,

$$\rho_{\text{star}} = \frac{3\pi}{GP^2} \left(\frac{a}{R_{\text{star}}} \right)^3.$$

This is a remarkable result: transit photometry alone, without any spectroscopic measurements, provides an estimate of the stellar density. This connects directly to stellar structure: the mean density constrains where the star sits in mass–radius space, complementing the spectral classification discussed above.



Schematic of an exoplanet transit showing the geometric configuration and the connection between observables and orbital parameters.



Single-transit observations of the hot Jupiter WASP-4b using NGTS (top) and WASP (bottom).
Credit: Wheatley et al. (2018).

In-Class Exercise: Transit Depth and Stellar Density

In this exercise, you will connect basic transit observables to physical properties of the star–planet system.

Assume:

- Orbital period $P = 3.0$ days
- Transit duration $t_T = 2.5$ hours
- Transit depth $\Delta F = 0.01$
- Circular orbit
- Edge-on geometry ($i \approx 90^\circ$, so $b \approx 0$)

Tasks

1. Compute the planet-to-star radius ratio $k = \sqrt{\Delta F}$.
2. Compute a/R_{star} and ρ_{star} .
3. Plot how ρ_{star} changes if t_T varies from 1 to 5 hours.
4. Compare your result to the solar mean density $\rho_{\odot} \approx 1.41 \text{ g cm}^{-3}$.

You may find the following useful:

$$\Delta F = k^2$$
$$\frac{a}{R_{\text{star}}} = \frac{1 + k}{\sin(\pi t_T / P)}$$
$$\rho_{\text{star}} = \frac{3\pi}{GP^2} \left(\frac{a}{R_{\text{star}}} \right)^3$$

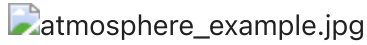
Use $G = 6.67 \times 10^{-8} \text{ cm}^3 \text{ g}^{-1} \text{ s}^{-2}$.

► Solution

Transmission Spectroscopy

Transmission spectroscopy observes a planet during transit to probe its atmosphere. This technique is the primary method for characterizing exoplanet atmospheres and directly connects stellar and planetary physics: the measurement requires precise knowledge of the host star's spectrum and limb darkening.

The time series of total brightness measurements is known as the transit light curve. During transit, the planet blocks a fraction of stellar flux equal to its projected area relative to the stellar disk. This fractional decrease is called the transit depth, δ .



Transit and eclipse geometry showing the planet radius R_{planet} , stellar radius R_{star} , atmospheric scale height H , and projected separation d . Credit: Robinson (2017).



Transit light curve of WASP-12b observed with the Hubble Space Telescope (Kreidberg et al. 2015).

Method

The key idea behind transmission spectroscopy is that the transit depth depends on wavelength.

At wavelengths where the atmosphere is more opaque due to atomic or molecular absorption, the effective radius of the planet increases slightly, leading to a deeper transit. The opacity sources in a planetary atmosphere are analogous to those in stellar atmospheres (Lecture 7): bound-bound transitions produce spectral features, while collision-induced absorption and Rayleigh scattering contribute to the continuum.

To measure this effect, the light curve is divided into spectrophotometric channels. Each channel is fit independently with a transit model, and the resulting wavelength-dependent transit depths form the transmission spectrum.



Near-infrared transmission spectrum of WASP-43b showing a prominent water absorption feature near $1.4 \mu\text{m}$.

To estimate the magnitude of spectral features, we define the atmospheric scale height,

$$H = \frac{k_B T_{\text{eq}}}{\mu g},$$

where T_{eq} is the planetary equilibrium temperature, μ is the mean molecular mass, and g is the surface gravity.

The amplitude of transmission features can be approximated as

$$\delta_\lambda \approx \frac{(R_{\text{planet}} + nH)^2 - R_{\text{planet}}^2}{R_{\text{star}}^2} \approx \frac{2nR_{\text{planet}}H}{R_{\text{star}}^2},$$

where n is the number of scale heights contributing to the signal, typically of order two for cloud-free atmospheres at low spectral resolution.

Even for favorable cases, the expected signal amplitude is only $\delta_\lambda \sim 10^{-3}$. For Earth-sized planets, the signal is two to three orders of magnitude smaller. The James Webb Space Telescope (JWST) has dramatically improved our ability to detect these signals, providing high-precision transmission spectra across the near- and mid-infrared for a growing sample of exoplanets.

In-Class Exercise: Transmission Spectroscopy Signal Size

In this exercise, you will estimate the expected amplitude of spectral features in transmission spectroscopy.

Assume a hot Jupiter with:

- $R_{\text{planet}} = 1.2R_J$
- $R_{\text{star}} = 1.0R_\odot$
- Equilibrium temperature $T_{\text{eq}} = 1500 \text{ K}$
- Mean molecular weight $\mu = 2.3m_p$
- Surface gravity $g = 15 \text{ m s}^{-2}$
- Number of scale heights probed $n = 2$

Tasks

1. Compute atmospheric scale height H .
2. Estimate δ_λ .
3. Plot δ_λ vs T_{eq} for T_{eq} from 500 to 2500 K.
4. Repeat for Earth-like planet ($R_{\text{planet}} = R_\oplus$, $g = 980 \text{ cm s}^{-2}$).

Useful equations:

$$H = \frac{k_B T_{\text{eq}}}{\mu g}$$

$$\delta_\lambda \approx \frac{2nR_{\text{planet}}H}{R_{\text{star}}^2}$$

Use:

- $k_B = 1.38 \times 10^{-16} \text{ erg K}^{-1}$
- $m_p = 1.67 \times 10^{-24} \text{ g}$

► Solution

Limb Darkening

Stars are not uniformly illuminated. Surface brightness varies due to stellar atmospheres, spots, faculae, plages, and flares. As a result, the apparent stellar limb appears dimmer than the center.

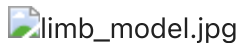
The physical origin of limb darkening is the variation of temperature with optical depth in the stellar atmosphere. When we look at the center of the stellar disk, our line of sight penetrates to deeper, hotter layers. Near the limb, the same optical depth is reached in shallower, cooler layers. Since the emitted intensity depends on local temperature (roughly following the Planck function), the limb appears dimmer. This connects directly to the radiative transfer and opacity physics discussed in Lecture 7: the opacity determines how deep we can see at each viewing angle.

This limb darkening introduces curvature into transit light curves and can modify the inferred transit depth by up to tens of percent.



Transit light curves of the same system using different limb-darkening prescriptions. All other parameters are held fixed.

Multiple limb-darkening models exist and must be constrained by stellar atmosphere calculations and observations. The simplest is a linear law, $I(\mu)/I_0 = 1 - u(1 - \mu)$, where $\mu = \cos \theta$ is the cosine of the angle from disk center. More realistic models use quadratic or nonlinear laws. The limb-darkening coefficients depend on the stellar effective temperature, surface gravity, and metallicity, and are typically computed from stellar atmosphere model grids.



Intensity maps from stellar atmosphere models at $T_{\text{eff}} = 5000$ K and $\log g = 0.0$ (Orosz & Hauschildt 2000).

In-Class Exercise: Limb Darkening and Transit Depth

In this exercise, you will explore how stellar limb darkening affects the transit light curve and inferred planet radius.

Assume:

- Circular, edge-on orbit ($i = 90^\circ$, $b = 0$)
- $R_{\text{planet}}/R_{\text{star}} = 0.1$
- Transit duration $t_T = 2.5$ hours
- Quadratic limb-darkening law:

$$\frac{I(\mu)}{I_0} = 1 - u_1(1 - \mu) - u_2(1 - \mu)^2, \quad \mu = \cos \theta$$

The transit depth at each time is approximated by:

$$\delta(t) = k^2 I(\mu(t))/I_0$$

and the normalized flux is

$$F(t) = 1 - \delta(t)$$

Tasks

1. Generate a simple transit light curve for three sets of limb-darkening coefficients:
 - Case A: no limb darkening $u_1 = u_2 = 0$
 - Case B: mild limb darkening $u_1 = 0.3, u_2 = 0.1$
 - Case C: strong limb darkening $u_1 = 0.6, u_2 = 0.3$
2. Plot all three curves on the same figure.
3. Comment on how the depth and shape of the transit changes.

► Solution

In-Class Assignment: Exploring Opacities in Stellar Models

In Lecture 7, we discussed how opacity governs energy transport inside stars. The total opacity κ determines whether energy flows primarily by radiation or convection, and its variation with density and temperature shapes the stellar structure. In this assignment, you will examine opacity profiles from MESA stellar models and compare them with analytic estimates from HKT Chapter 4.

Recall that the effective opacity combines radiative and conductive contributions as

$1/\kappa = 1/\kappa_{\text{rad}} + 1/\kappa_{\text{cond}}$, so the smaller opacity (more efficient transport mechanism) dominates.

Your favorite model files are again:

- $1 M_{\odot}$: [M1_default_profile8.data](#); [M1_default_profile218.data](#);
- $15 M_{\odot}$: [M15_aprox21_profile8.data](#); [M15_aprox21_profile19.data](#);

κ_{rad}

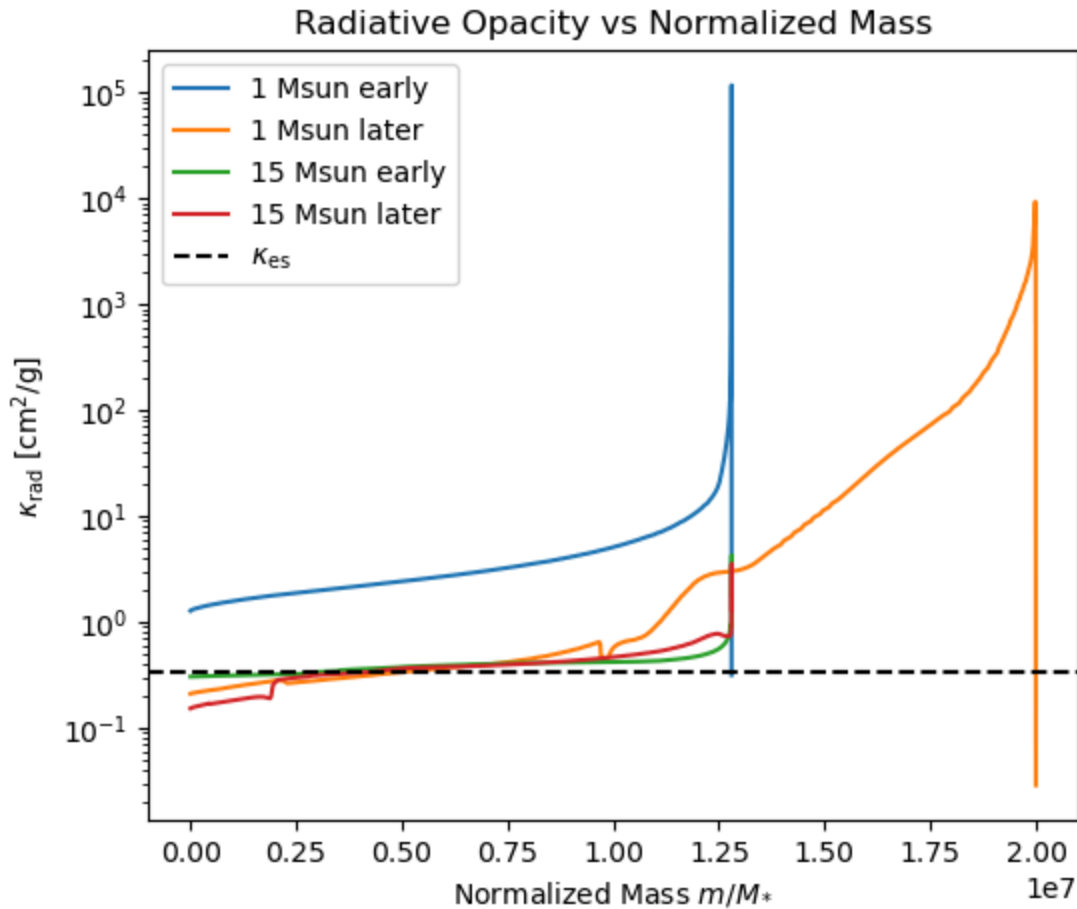
Using the $1 M_{\odot}$ and $15 M_{\odot}$ MESA models at their early evolutionary phase, plot the opacity. Consider using the mass scale in Figure 4.6 of HKT to emphasize the outer layers.

Hint: this requires you to normalize by the total mass of the star!

On the same plot, plot an estimate of the κ_{rad} of your choosing assuming a uniform mixture. An example of this is on HKT 216. Do the opacity profiles match your estimated value? Which opacity source did you choose and why?

Some useful analytic estimates from HKT Chapter 4:

- Electron scattering (Thomson): $\kappa_{\text{es}} = 0.2(1 + X) \approx 0.34 \text{ cm}^2 \text{ g}^{-1}$ for solar composition. This is independent of temperature and density.
- Kramers opacity (bound-free and free-free): $\kappa_{\text{bf+ff}} \propto \rho T^{-3.5}$, which increases toward the cooler, lower-density outer layers.
- In the deep interior of massive stars, electron scattering typically dominates. In the outer layers, bound-free and free-free opacities rise sharply, and molecular opacities contribute for cooler stars.



κ_{cond}

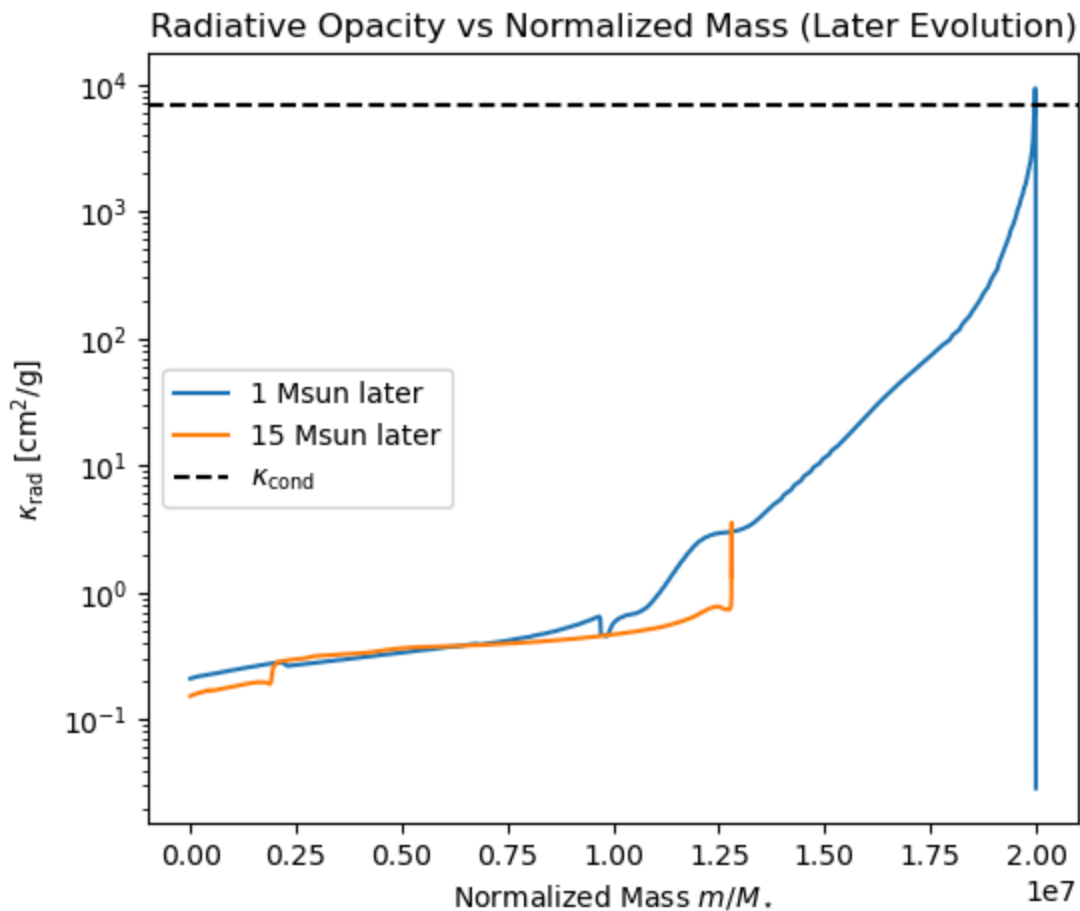
Perform the same exercise here for the *later* evolutionary profiles. This time, compare with an estimate for the κ_{cond} in HKT 4.72. Do the profiles match? Why?

At later evolutionary stages, stellar cores become denser and may develop electron degeneracy. In degenerate matter, electrons have long mean free paths and conduct heat very efficiently, giving a small κ_{cond} . Since the total opacity combines as $1/\kappa = 1/\kappa_{\text{rad}} + 1/\kappa_{\text{cond}}$, conduction dominates (smaller κ) wherever $\kappa_{\text{cond}} \ll \kappa_{\text{rad}}$. This typically occurs in the dense, degenerate cores of evolved stars and in white dwarfs (Lecture 8), where conduction is so efficient that the interior is nearly isothermal.

```

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taxWarning: invalid escape sequence '\s'
plt.xlabel('Normalized Mass $m/M_\star$')

```



Observations

- In later stages, the outer layers may have higher densities or degenerate regions, so conductive opacity can become significant.
- Comparing the MESA profile with κ_{cond} shows that in these outer layers, $\kappa_{\text{rad}} \gg \kappa_{\text{cond}}$, so conduction is still subdominant.
- Deeper inside the star (high ρ , partially degenerate core), κ_{cond} may dominate over radiative opacity.
- This demonstrates that different opacity sources dominate in different regions depending on density, temperature, and degeneracy.

In-Class Assignment: Exploring Brown Dwarfs and Planets with MESA

This assignment takes us to the low-mass end of the stellar spectrum, where objects are too low in mass to sustain hydrogen fusion. In Lecture 7, we discussed the nuclear burning thresholds set by the Gamow peak. Brown dwarfs fall below the hydrogen-burning limit but above the deuterium-burning limit, while planets fall below both. In Lecture 8, we discussed star formation through Jeans collapse; planets form in the circumstellar disks that are a byproduct of this collapse process.

Here we use MESA models to explore the internal structure, opacity, and evolution of brown dwarfs and giant planets.

Download the following model files locally.

- $0.03 M_{\odot}/1 R_{\text{Jup}}$: [0.03Msun_1RJup_history.data](#); [0.03Msun_1RJup_profile.data](#);
- $0.97 M_{\text{Jup}}/2 R_{\text{Jup}}$: [planet_1.0_MJ_10.0_ME_2.0_RJ_0_irr_flux.data](#);
[planet_1.0_MJ_10.0_ME_2.0_RJ_1e6_irr_flux.data](#);
[planet_1.0_MJ_10.0_ME_2.0_RJ_1e8_irr_flux.data](#);

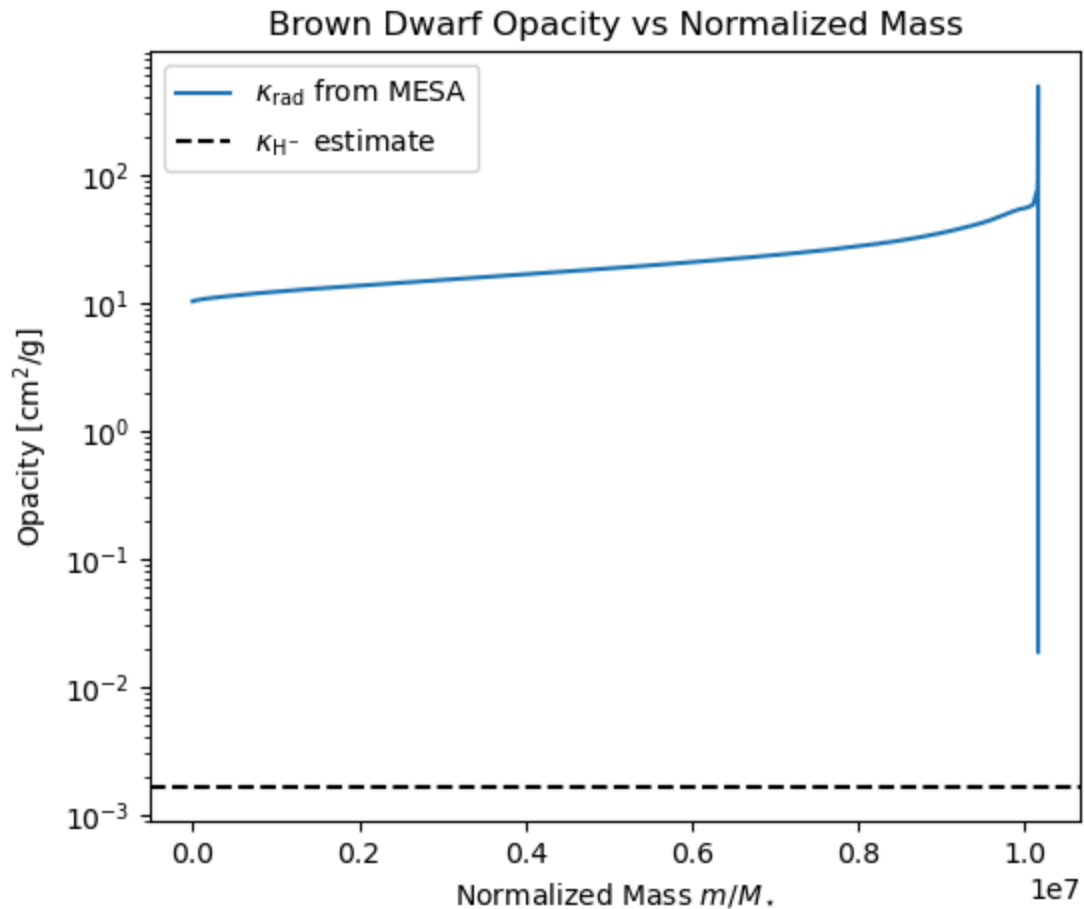
Brown Dwarf

Using the $0.03 M_{\odot}$ model MESA profile:

a.

Compare the `opacity` profile with the approximate opacity due to H^- from HKT 4.65. The H^- ion (a hydrogen atom with an extra electron) is the dominant continuum opacity source in cool stellar and substellar atmospheres. Its bound-free absorption produces the opacity floor that makes the Sun and cooler objects opaque in the visible.

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ntaxWarning: invalid escape sequence '\s'
plt.xlabel('Normalized Mass $m/M_{\star}$')
```

Discussion

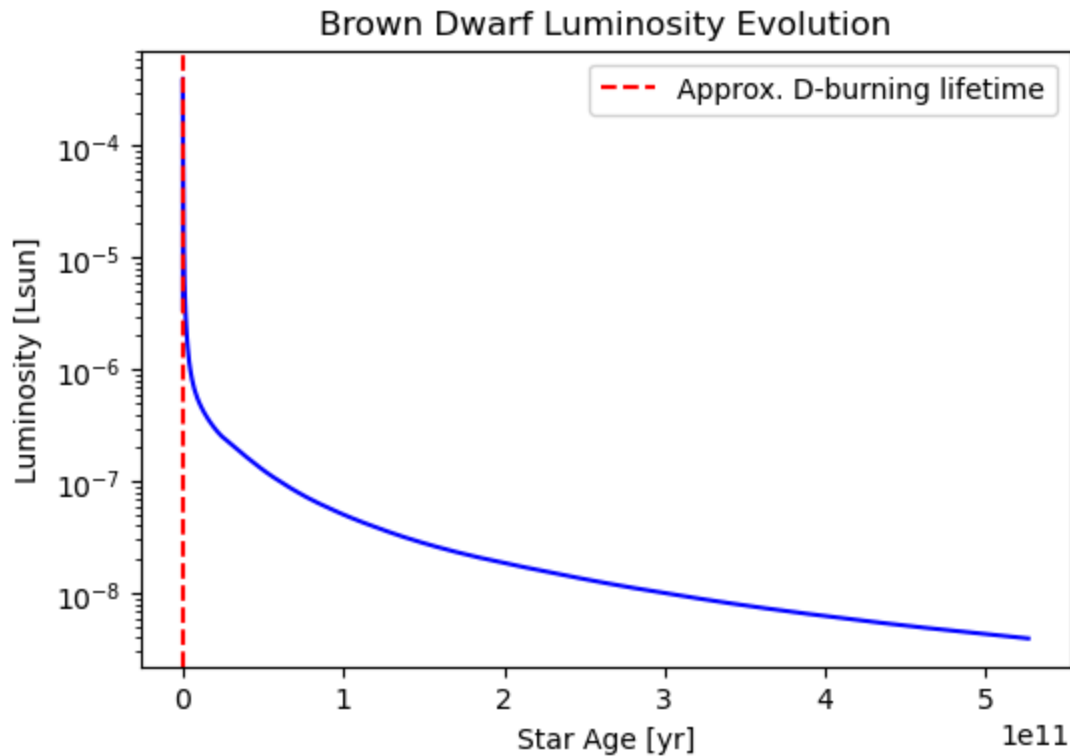
- The H^- opacity estimate is much smaller ($\sim 10^{-3} \text{ cm}^2/\text{g}$) than the total opacity from MESA ($\sim 10 \text{ cm}^2/\text{g}$).
- This is expected because H^- is just one contribution.
- MESA includes molecular opacities, bound-free, free-free, and H^- , so the total opacity is much higher.

b.

Using the dominant nuclear energy generation rate (ϵ), luminosity, and other necessary variables, compute the estimated lifetime from HKT eq. 1.89. Consider plotting ^2H as a function of time. How far through the star's lifetime is this model?

Recall from Lecture 7 that deuterium burning has a lower temperature threshold than hydrogen burning because the Coulomb barrier for $^2\text{H} + p$ is smaller. Brown dwarfs burn through their deuterium supply in roughly tens of millions of years, after which they simply cool and fade.

Estimated deuterium-burning lifetime: $2.17 \times 10^7 \text{ yr}$



Planet

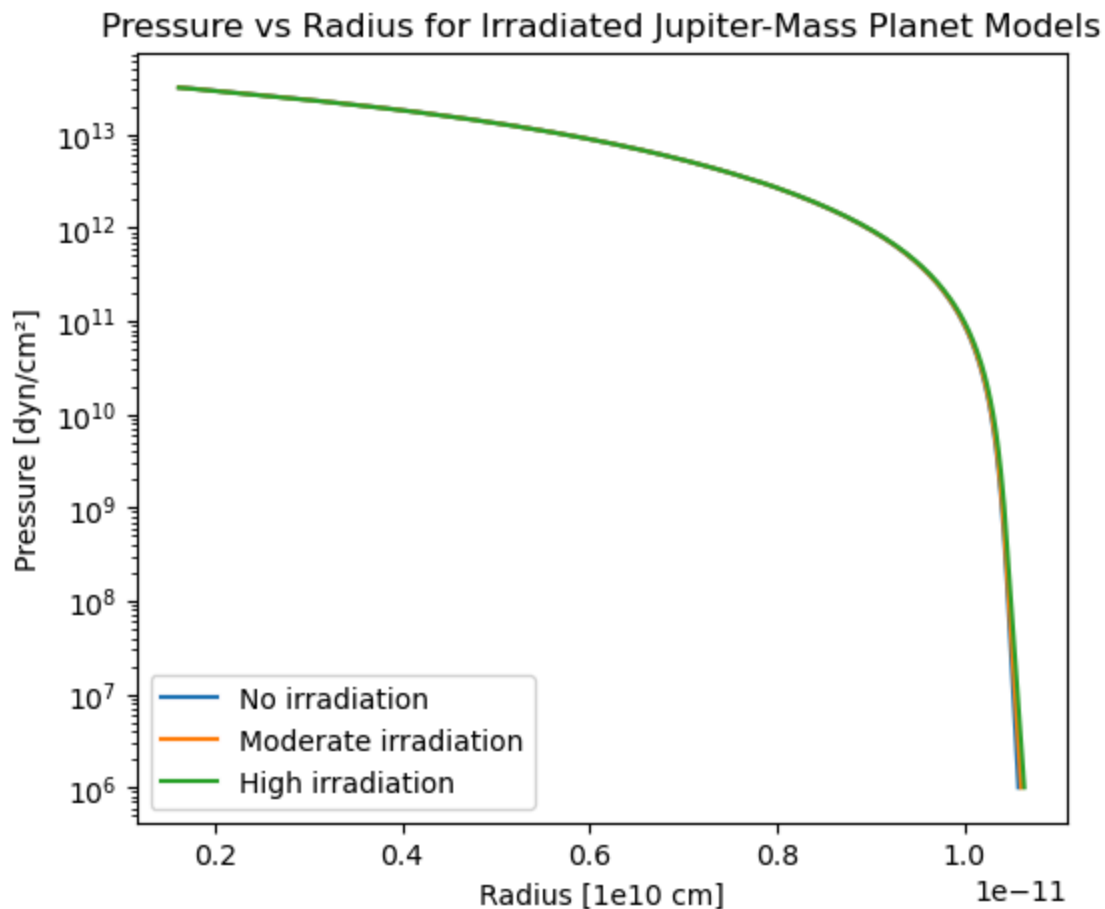
Using the $0.97 M_{\text{Jup}}/2 R_{\text{Jup}}$ model MESA history data:

Each of these models uses a set value of irradiation flux of 0, 1×10^6 , or $1 \times 10^8 \text{ erg cm}^{-2} \text{ s}^{-1}$ at a fixed column density of $300 \text{ cm}^2 \text{ g}^{-1}$.

C.

Plot the pressure vs radius for all 3 models. Which model has the largest radius? Why?

Irradiation from a close-in host star heats the planet's outer atmosphere, inflating the envelope. This is one proposed explanation for the observed radius inflation of hot Jupiters, whose radii often exceed predictions from cooling models alone.



d.

A shallower atmospheric temperature gradient leads to slower interior cooling and a larger radius at a given age.

Plot the pressure vs temperature (x-axis) profiles for all 3 models. Which model has the flattest temperature gradient?

The connection between atmospheric heating and radius is subtle: irradiation flattens the outer temperature profile, which reduces the temperature contrast driving convective and radiative heat transport from the interior. Slower cooling means the planet retains more thermal energy and therefore stays inflated for longer.

Pressure vs Temperature for Irradiated Jupiter-Mass Planet Models

